Module 4 — Quiz A (Def & Notation)

Practice bank, matches the Canvas quiz, open notes, unlimited attempts

Module 4, Quiz A: Definitions, notation, and two-group methods

Reading: Module 4 Read 1. Project: Project 4.

These practice questions match the autograded Canvas quiz. Each question is self-contained: all data, names, and context you need are in the question. Work them after Read 1, then check the answer key at the bottom. Topics follow Read 1: study-design vocabulary and causal language (Section 1), choosing and reading two-group tests and confidence intervals (Section 2), comparing more than two groups, and the reasoning behind simulating data.

For integrated scenarios that combine design, test choice, R code, and graph reading, see the Synthesis quiz.

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1. Explanatory variable

In a two-group comparison, which best defines the explanatory variable (EV)?

- A. Any variable measured on both groups, regardless of role.
- B. The response variable we measure to assess an effect.
- C. The variable thought to explain or cause changes in the response; the treatment, exposure, or grouping factor.
- D. A variable that must always be numerical.

2. Response variable

In a two-group comparison, which best defines the response variable (RV)?

- A. The grouping variable, treatment versus control.
- B. The variable that is randomly assigned in every study.
- C. The variable we measure to see whether it changes with the explanatory variable.
- D. A variable that always has a normal distribution.

3. Observational study

Which best defines an observational study?

- A. Researchers randomly assign participants to treatment or control groups.
- B. Researchers record data as they are; group membership is set by the participants, not by random assignment.
- C. Any study that has both a control group and a treatment group.
- D. A study that always collects a full census of the population.

4. RCT key feature

What single feature most distinguishes a randomized controlled trial (RCT) from an observational study?

- A. A large sample size (n greater than 100).
- B. Collecting data over many years.
- C. Measuring both a numerical and a binary outcome.
- D. Random assignment of units to groups by the researcher.

5. What random assignment does

What does random assignment accomplish in a study?

- A. It removes all measurement error.
- B. It guarantees equal group sizes.
- C. It balances other variables, measured and unmeasured, across the groups on average, removing confounding as an explanation.
- D. It guarantees the sample represents the national population.

6. What random sampling does

How is random sampling different from random assignment?

- A. Random sampling is only used in experiments.
- B. Random sampling controls confounding; random assignment controls generalization.
- C. Random sampling selects individuals from a population to reduce bias when generalizing; random assignment splits a sample into comparison groups to control confounding.
- D. They are two names for the same procedure.

7. Confounder definition

Which best defines a confounder for the effect of an explanatory variable on a response?

- A. A variable that only affects the response variable.
- B. Any variable measured in a study beyond the EV and RV.
- C. The random error term in a model.
- D. A variable that influences both the explanatory variable and the response, distorting the apparent relationship between them.

8. Confounder in a DAG

In a causal DAG, which arrows define a confounder C for the effect of group X on outcome Y?

- A. One arrow from C to X and another arrow from C to Y.
- B. A single arrow from X to Y only.
- C. One arrow from X to C and one from Y to C.
- D. An arrow from Y to X.

9. What a DAG represents

In a causal directed acyclic graph (DAG), what do the nodes and directed edges represent?

- A. Nodes are p-values; edges are confidence intervals.
- B. Nodes are data values; edges are correlation coefficients.
- C. Nodes are groups; edges are sample sizes.
- D. Nodes are variables; a directed edge from A to B means A is assumed to help cause B.

10. Backdoor path

A study compares two universities (X) on calculus passing (Y). Students who took AP calculus (C) are both more likely to enroll at one university and more likely to pass. What is the non-causal X to Y route through C called, and how do you block it?

- A. A collider; block it by ignoring C entirely.
- B. A direct causal path; block it by increasing the sample size.
- C. A backdoor path; block it by subclassifying (comparing universities within AP and no-AP groups).
- D. A mediator path; block it by removing Y.

11. Subclassification limitation

You compare two groups within levels of a measured confounder (subclassification). What is its main limitation?

- A. It requires the response variable to be binary.

- B. It always increases the p-value.
- C. It removes the need for any study design.
- D. It can only adjust for confounders you actually measured; unmeasured confounding can remain.

12. Internal validity

Which best defines internal validity?

- A. How well study results generalize to a larger population.
- B. How well a study rules out alternative explanations for a difference between its own groups.
- C. The number of variables measured.
- D. The size of the p-value.

13. Simpson's paradox

University A has a lower overall calculus pass rate than University B, yet A has a higher pass rate within both the AP-calculus group and the no-AP group. What is this reversal called?

- A. Simpson's paradox, caused by a confounder (AP background) distributed unevenly across universities.
- B. A Type I error.
- C. Random sampling error that disappears with more data.
- D. A mediator effect.

14. Three explanations for a difference

Two groups in a dataset differ on the response variable. Before claiming a treatment effect, which three explanations should you always consider?

- A. Type I, Type II, and Type III errors.
- B. Sampling, rounding, and typing errors.
- C. Mean, median, and mode.
- D. By chance, causation, and confounding.

15. p-value does not prove causation

In an observational study, a very small p-value for a two-group difference by itself establishes that the explanatory variable caused the difference in the response.

- A. True
- B. False

16. Identify design: smoking

Researchers compare lung-cancer rates between adults who smoke and adults who do not, using survey records. No one was assigned to smoke. What is the design and its consequence?

- A. Observational; a difference still proves causation if the p-value is small.
- B. RCT; smoking was the randomly assigned treatment.
- C. Observational; a difference shows association, not necessarily causation, because smokers may differ in other ways.
- D. RCT; a difference shows causation because the sample is large.

17. Identify design: exercise trial

Sixty volunteers are randomly assigned to an 8-week aerobic program or a stretching program, and resting heart rate is measured. What is the design and its consequence?

- A. RCT; because of random assignment, a significant difference supports a causal conclusion.
- B. Observational; only association can be claimed.
- C. Observational; participants chose their program.

- D. RCT; but no causal claim is possible because n is only 60.

18. Consequence of an RCT result

An RCT randomly assigns 120 patients to drug or placebo; the drug group has significantly lower blood pressure. What can be concluded?

- A. The drug likely caused the lower blood pressure, because randomization removed confounding.
- B. Only that drug use is associated with lower blood pressure.
- C. The drug caused the effect only if the sample exceeds 200.
- D. Nothing, because observational studies cannot show causation.

19. Consequence of an observational result

An observational survey finds standing-desk users report 15 percent higher productivity, with a small p-value. What is the strongest valid conclusion?

- A. Nothing can be concluded from observational data.
- B. Standing desks cause higher productivity, because the p-value is small.
- C. Standing desks cause higher productivity only for this sample.
- D. Standing-desk use is associated with higher productivity; unmeasured confounders could explain the gap.

20. Identify EV and RV

A study asks whether a new tutoring program raises final-exam scores. Students are assigned to tutoring or no tutoring, and their exam scores are recorded. Which is the explanatory variable and which is the response?

- A. EV: student name; RV: tutoring group.
- B. EV: final-exam score; RV: tutoring group.
- C. EV: tutoring group (tutoring vs none); RV: final-exam score.
- D. EV and RV are both the exam score.

21. Parameter for a numerical outcome

You compare a numerical outcome (reaction time in ms) between two independent groups. What is the parameter of interest?

- A. The correlation between the two groups.
- B. A difference in means, mu - mu .
- C. A difference in proportions, p - p .
- D. The slope of a regression line.

22. Parameter for a binary outcome

You compare a binary outcome (recovered: yes or no) between two independent groups. What is the parameter of interest?

- A. The standard deviation of the response.
- B. A difference in means, mu - mu .
- C. The median difference.
- D. A difference in proportions, p - p .

23. Name the parameter: cholesterol

A trial compares mean LDL cholesterol (mg/dL) between a drug group and a placebo group. Which parameter and summary fit?

- A. Difference in proportions p - p ; summarize with group rates.
- B. A single proportion; summarize with one percentage.
- C. A correlation; summarize with a scatterplot only.

- D. Difference in means mudrug - mu lacebo; summarize each group with mean and SD.

24. Name the parameter: infection

A trial compares whether each participant got the flu (yes or no) between a vaccine group and a placebo group. Which parameter and summary fit?

- A. An IQR difference; summarize with boxplots.
- B. A slope; summarize with a regression line.
- C. Difference in means mu - mu ; summarize with group means.
- D. Difference in proportions pvaccine - p lacebo; summarize with group infection rates.

25. Choose the test: numerical, two groups

Two independent groups are compared on a numerical outcome (blood pressure). Which test and R call fit?

- A. Two-proportion test: prop.test(c(x1, x2), c(n1, n2)).
- B. Welch two-sample t-test: t.test(bp ~ group, data = df, var.equal = FALSE).
- C. Chi-square test: chisq.test(tab).
- D. Paired t-test: t.test(x, y, paired = TRUE).

26. Choose the test: binary, two groups

Two independent groups are compared on a binary outcome (recovered: yes or no), with 80 and 75 people. Which test fits?

- A. Welch two-sample t-test: t.test(outcome ~ group).
- B. Paired t-test on the two counts.
- C. One-sample binomial test: binom.test(x1, 80).
- D. Two-proportion test or chi-square test: prop.test(c(x1, x2), c(80, 75)) or chisq.test(tab).

27. Conditions for the two-sample t-test

Which conditions does the methods map list for a Welch two-sample t-test comparing a numerical outcome across two groups?

- A. The two measurements come from the same units (paired).
- B. All expected cell counts at least 5.
- C. Equal sample sizes in both groups.
- D. Two independent samples (or random assignment), and each group large or roughly normal/symmetric.

28. Conditions for the chi-square test

Which condition does the methods map list for a chi-square test of independence on a 2x2 table?

- A. The response variable is numerical and roughly normal.
- B. The data are paired before-and-after measurements.
- C. Independent observations and all expected counts at least 5.
- D. The two groups have equal variances.

29. Best plot: numerical, two groups

Which plot best compares a numerical outcome across two groups?

- A. A boxplot per group (side-by-side).
- B. A 100 percent stacked bar.
- C. A scatterplot of group versus group.
- D. A single pie chart.

30. Best plot: binary, two groups

Which plot best compares a binary outcome (event: yes or no) across two groups?

- A. A boxplot per group.
- B. A 100 percent stacked bar of the response variable by group, using geom_bar(position = ‘fill’).
- C. A scatterplot.
- D. A histogram of the response variable.

31. Why Welch by default

Why does R’s t.test() use the Welch approximation by default?

- A. Welch does not assume equal population variances, so it stays valid when the two groups have different spreads.
- B. Welch only works for binary outcomes.
- C. Welch always gives a smaller p-value.
- D. Welch requires a larger sample size to run.

32. Read a t-test result

A Welch two-sample t-test gives p = 0.03 with a 95 percent CI for mu_B - mu_A of [1.2, 8.6]. At = 0.05, what do you conclude?

- A. Cannot decide without the sample sizes.
- B. Fail to reject H , because p > 0.05.
- C. Reject H ; the data support a higher mean for group B, by about 1.2 to 8.6 units.
- D. Reject because the interval is wide.

33. Read a proportion-test result

A two-proportion test gives p = 0.18 with a 95 percent CI for p - p of [-0.02, 0.10]. At = 0.05, what do you conclude?

- A. Cannot decide without the raw counts.
- B. Fail to reject H ; the interval includes 0, so there is not enough evidence of a difference.
- C. Reject H ; the interval is entirely positive.
- D. Reject because 0.18 is near 0.05.

34. Null hypothesis for two means

You test whether two independent population means differ. Which null hypothesis is correct?

- A. H : p = p .
- B. H : x̄_1 = x̄_2 (a claim about sample means).
- C. H : mu > mu .
- D. H : mu = mu , equivalently mu - mu = 0.

35. Choose the test: paired numerical

Each of 25 runners has resting heart rate measured before and after a 10-week program (the same runners are measured twice). Which test compares before and after?

- A. Paired t-test on the within-runner differences: t.test(before, after, paired = TRUE).
- B. Chi-square test: chisq.test(tab).
- C. Welch two-sample t-test: t.test(rate ~ time).
- D. Two-proportion test: prop.test(…).

36. Choose the test: paired binary

The same 100 voters answer support or do-not-support before and after a debate. Which test compares the before and after support rates?

- A. McNemar's test: mcnemar.test(tab), because the two yes/no answers are paired.
- B. Welch two-sample t-test on support.
- C. One-way ANOVA across the two time points.
- D. Two-proportion test: prop.test() on the two counts.

37. Paired or independent: men vs women

Group 1 is 40 men and group 2 is 40 different women; you compare mean income. Are the data paired or independent?

- A. Independent; the two groups are separate, unrelated people.
- B. Paired; the sample sizes are equal.
- C. Paired; both groups are people.
- D. Independent; only if incomes are normal.

38. Paired or independent: pre/post

Each of 30 students takes a pre-test and a post-test; you compare mean scores. Are the data paired or independent?

- A. Paired; the same student contributes both scores.
- B. Independent; pre and post are different tests.
- C. Independent; there are two score columns.
- D. Paired only if the class is large.

39. Paired or independent: two arms

For 50 patients, blood pressure is measured on the left arm and the right arm; you compare mean readings. Are the data paired or independent?

- A. Independent; left and right arms are different.
- B. Paired only if readings are equal.
- C. Independent; there are 100 readings total.
- D. Paired; both readings come from the same patient.

40. Purpose of set.seed()

Before generating random data in a simulation you write set.seed(42). What does this accomplish?

- A. It guarantees the p-value will be below 0.05.
- B. It fixes the random-number generator so the results are reproducible; anyone running the code gets the same numbers.
- C. It makes the simulated sample larger and more accurate.
- D. It removes confounding from the simulated data.

41. p-values under the null

You simulate many datasets in which two groups have the same true mean, and you record each t-test p-value. What should you expect?

- A. Almost all p-values fall below 0.05.
- B. The p-values are roughly uniform on 0 to 1, and about 5 percent fall below 0.05.
- C. Every p-value equals exactly 0.05.
- D. The p-values cluster tightly around 0.5.

42. By chance? numerical, clear gap

The boxplots show a numerical outcome for two groups of 40. The boxes barely overlap and the treatment median is far above the control median. Does the difference look like it could easily be due to chance?

- A. Cannot tell, because boxplots never show group differences.
- B. Yes; any difference between two groups is always just chance.
- C. Yes; the boxes are the same height.
- D. No; the clear separation with 40 per group is unlikely to be chance, so a small p-value would be expected.

43. By chance? numerical, heavy overlap

The boxplots show a numerical outcome for two groups of 40. The two boxes overlap almost completely and the medians are nearly the same. Does the difference look like it could easily be due to chance?

- A. No; overlap proves the groups differ.
- B. Cannot tell; boxplots cannot show overlap.
- C. Yes; heavily overlapping boxes with similar medians are consistent with chance, so a large p-value would be expected.
- D. No; the treatment clearly works.

44. By chance? binary, clear difference

The 100 percent stacked bars show the event rate for two groups of 100. The control event rate is about 0.70 and the treatment event rate is about 0.30. Does the difference in rates look like it could easily be due to chance?

- A. Cannot tell from a stacked bar.
- B. No; a 40 percentage-point gap with 100 per group is unlikely to be chance, so a small p-value would be expected.
- C. Yes; the bars are the same height.
- D. Yes; proportions always differ by chance.

45. By chance? binary, similar rates

The 100 percent stacked bars show the event rate for two groups of 100. The two event rates are about 0.52 and 0.47. Does the difference in rates look like it could easily be due to chance?

- A. Cannot tell; rates cannot be compared.
- B. No; stacked bars always show real effects.
- C. No; any difference proves the treatment works.
- D. Yes; a 5 percentage-point gap with this sample size is easily consistent with chance, so a large p-value would be expected.

Answer key

1. Correct answer: C. The EV is the treatment, exposure, or group label whose effect you study.
2. Correct answer: C. The RV is the response variable you compare across groups.
3. Correct answer: B. In an observational study, group membership is self-selected, so confounding is a live concern.
4. Correct answer: D. Random assignment is the defining feature; it balances confounders across groups.
5. Correct answer: C. Random assignment balances confounders across groups, so a difference is not due to confounding.
6. Correct answer: C. Sampling is about who is in the study (generalizing); assignment is about which group they go into (confounding).
7. Correct answer: D. A confounder is a common influence on both the EV and the RV.
8. Correct answer: A. A confounder has arrows pointing out to both the group and the outcome.
9. Correct answer: D. A DAG encodes assumed causal directions between variables.
10. Correct answer: C. A common cause opens a backdoor path; conditioning on (subclassifying by) it blocks the path.
11. Correct answer: D. Subclassification handles measured confounders only; unmeasured ones can still bias the comparison.
12. Correct answer: B. Internal validity is about ruling out alternative explanations within the study.
13. Correct answer: A. When a crude comparison reverses after accounting for a confounder, that is Simpson's paradox.
14. Correct answer: D. A difference could be chance, a real causal effect, or confounding; design and a test sort these out.
15. Correct answer: B (False). Correct. A small p-value only rules out chance; confounding can still explain an observational difference.
16. Correct answer: C. Smoking is self-selected, so this is observational and supports association only.
17. Correct answer: A. Random assignment makes this an RCT, so a significant difference supports causation.
18. Correct answer: A. In an RCT, a significant difference supports a causal conclusion.
19. Correct answer: D. Observational data support association; a small p-value does not remove confounding.
20. Correct answer: C. The group label (tutoring vs none) is the EV; the measured exam score is the RV.
21. Correct answer: B. A numerical outcome compares means, so the parameter is a difference in means.
22. Correct answer: D. A binary outcome compares proportions (rates), so the parameter is a difference in proportions.
23. Correct answer: D. Cholesterol is numerical, so compare means and summarize with means and SDs.
24. Correct answer: D. Flu yes/no is binary, so compare infection proportions (rates).
25. Correct answer: B. A numerical outcome across two independent groups uses the Welch two-sample t-test.
26. Correct answer: D. A binary outcome across two independent groups uses prop.test or chisq.test.
27. Correct answer: D. Independence (or random assignment) plus approximate normality or large n are the t-test conditions.
28. Correct answer: C. Chi-square needs independent observations and all expected counts at least 5.
29. Correct answer: A. Side-by-side boxplots compare the center and spread of a numerical outcome across groups.
30. Correct answer: B. A 100 percent stacked bar shows each group's event proportion for easy comparison.
31. Correct answer: A. Welch does not require equal variances, making it the safer default.
32. Correct answer: C. p below 0.05 and a CI excluding 0 both point to a real difference (B higher).
33. Correct answer: B. p above 0.05 and a CI containing 0 mean you fail to reject the null.
34. Correct answer: D. The null is about population means being equal, not sample means.
35. Correct answer: A. The same runners measured twice are paired, so use the paired t-test on the differences.
36. Correct answer: A. Paired binary outcomes (same people twice) use McNemar's test.
37. Correct answer: A. Different people in each group means the samples are independent.
38. Correct answer: A. The same students measured twice gives paired data.
39. Correct answer: D. Two measurements on the same patient are paired.
40. Correct answer: B. set.seed fixes the generator state for reproducible results.
41. Correct answer: B. Under a true null, p-values are uniform, so about 5 percent are below 0.05 (the Type I error rate).
42. Correct answer: D. A large separation relative to spread, with decent n, is unlikely under chance alone.
43. Correct answer: C. When boxes overlap heavily and medians are close, the difference is plausibly chance (expect a large p-value).

44. Correct answer: B. A large rate gap (0.70 vs 0.30) with $n = 100$ per group is unlikely under chance.
45. Correct answer: D. A small rate gap (0.52 vs 0.47) is easily produced by chance; expect a large p-value.